

What is Service in Angular



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Disadvantages

1. Same code is repeated in three different component classes. We are violating DRY principal.
2. Component class should only be responsible for representing UI to the user.
3. Presentation logic must be separated from business logic to make components more maintainable.

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Advantages

1. Services allows us to re-use a piece of code in multiple components, wherever it is required. In this way we avoid repeating a piece of code.
2. It allows us to separate business logic from UI logic. We write UI logic in component class and business logic in a service class. In this way it provides separation of concern.
3. We can unit test the business logic written in a service class separately without creating a component. Testing and debugging is easier with services.

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A service in Angular is a re-usable TypeScript class that can be used in multiple components across our Angular application.



We can also use services to communicate between two non-related components in Angular



Dependency Injection (DI)



A **dependency is a relationship between two software components where one component relies on the other component to work properly.**



Disadvantage of not using Dependency Injection (DI)

1. Without dependency injection, a class is tightly coupled with its dependency. This makes a class non-flexible. Any change in dependency forces us to change the class implementation.
2. It makes testing of class difficult. Because if the dependency changes, the class has to change. And when the class changes, the unit test mock code also has to change.



```
@Component({
  selector: 'app-header',
  template: 'header.component.html',
  provider: [SubscribeService]
})
export class HeaderComponent{
  subService: SubscribeService;

  constructor(subService: SubscribeService){
    this.subservice = subservice;
  }

  OnSubscribe(){
    this.SubService.OnSubscribeClicked();
  }
}
```

Injector



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angular-services-dependency-injection

EXPLORER

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ANGULAR-SERVICES-DEPENDENCY-...

hero.component.html

TS hero.component.ts

▼ sidebar

◀ sidebar.component.html

TS sidebar.component.ts

◀ home.component.html

TS home.component.ts

◀ header.component.html

TS header.component.ts

▼ Services

TS subscribe.service.ts

app.component.css

◀ app.component.html

TS app.component.spec.ts

TS app.component.ts

TS app.module.ts

> assets

★ favicon.ico

◀ index.html

TS main.ts

TS subscribe.service.ts

TS header.component.ts

TS hero.component.ts

TS sidebar.component.ts

src > app > header > home > sidebar > TS sidebar.component.ts > SidebarComponent

1 import { Component } from '@angular/core';

2 import { SubscribeService } from '../../Services/subscribe.service';

3

4 @Component({

5 selector: 'app-sidebar',

6 templateUrl: './sidebar.component.html',

7 providers: [SubscribeService] //2. WHAT TO PROVIDE

8 })

9 export class SidebarComponent {

10

11 //1. HOW TO PROVIDE DEPENDENCY

12 constructor(private subService: SubscribeService){

13

14 }

15

16 OnSubscribe(){

17 this.subService.OnSubscribeClicked('quarterly');

18 }

19 }

20



Dependency injection (DI) is a technique (design pattern) using which a class receives its dependencies from an external source rather than creating them itself.

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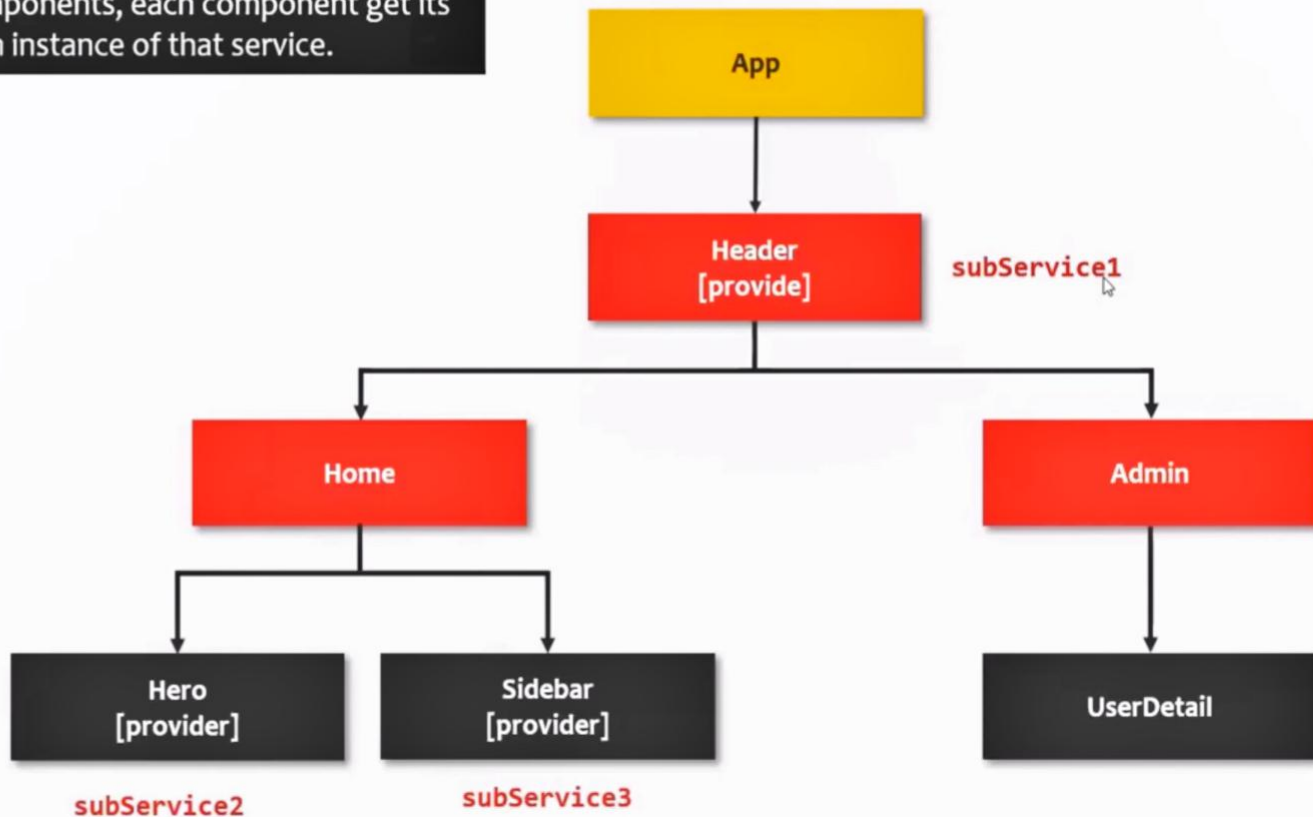
Advantage of using Dependency Injection (DI)

1. Dependency injection or DI keeps the code flexible, testable, and mutable
2. Classes can inherit external logic without knowing how to create it.
3. Dependency injection benefits components, directives and pipes.



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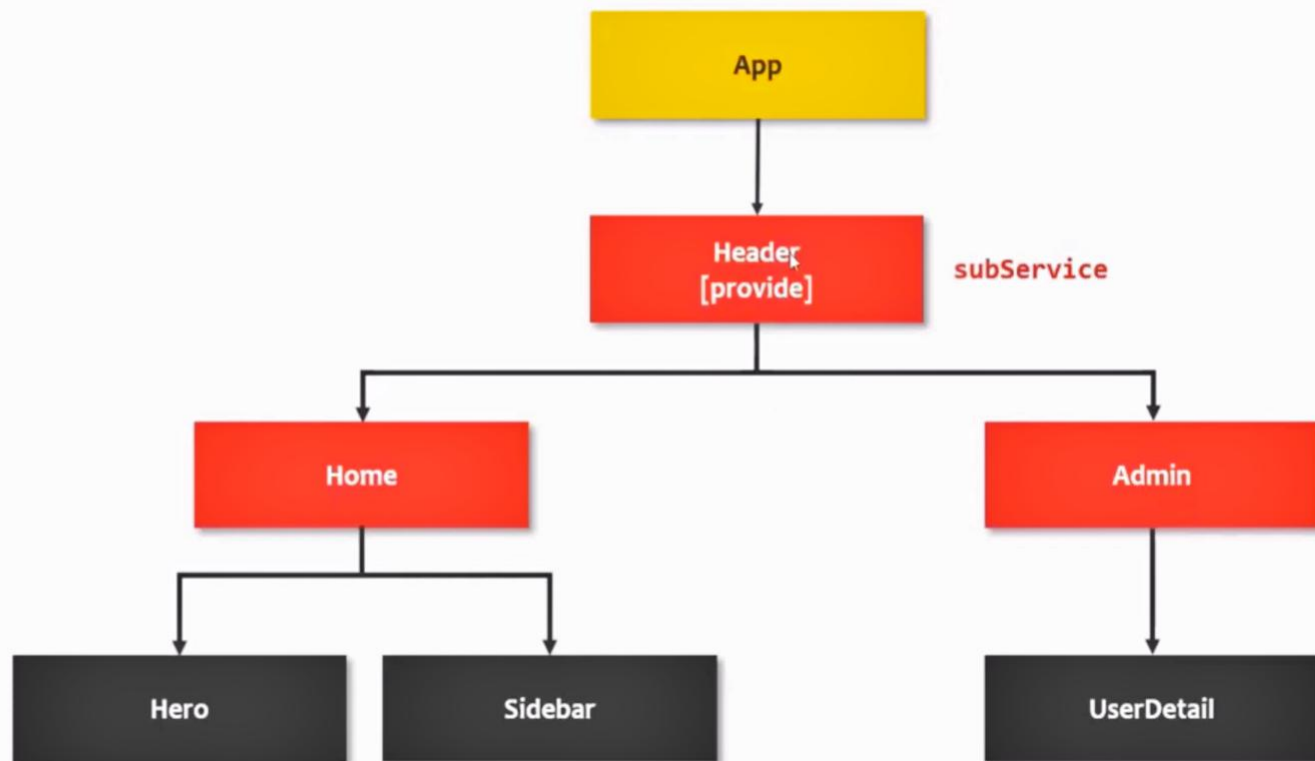
When we provide a service on multiple components, each component get its own instance of that service.

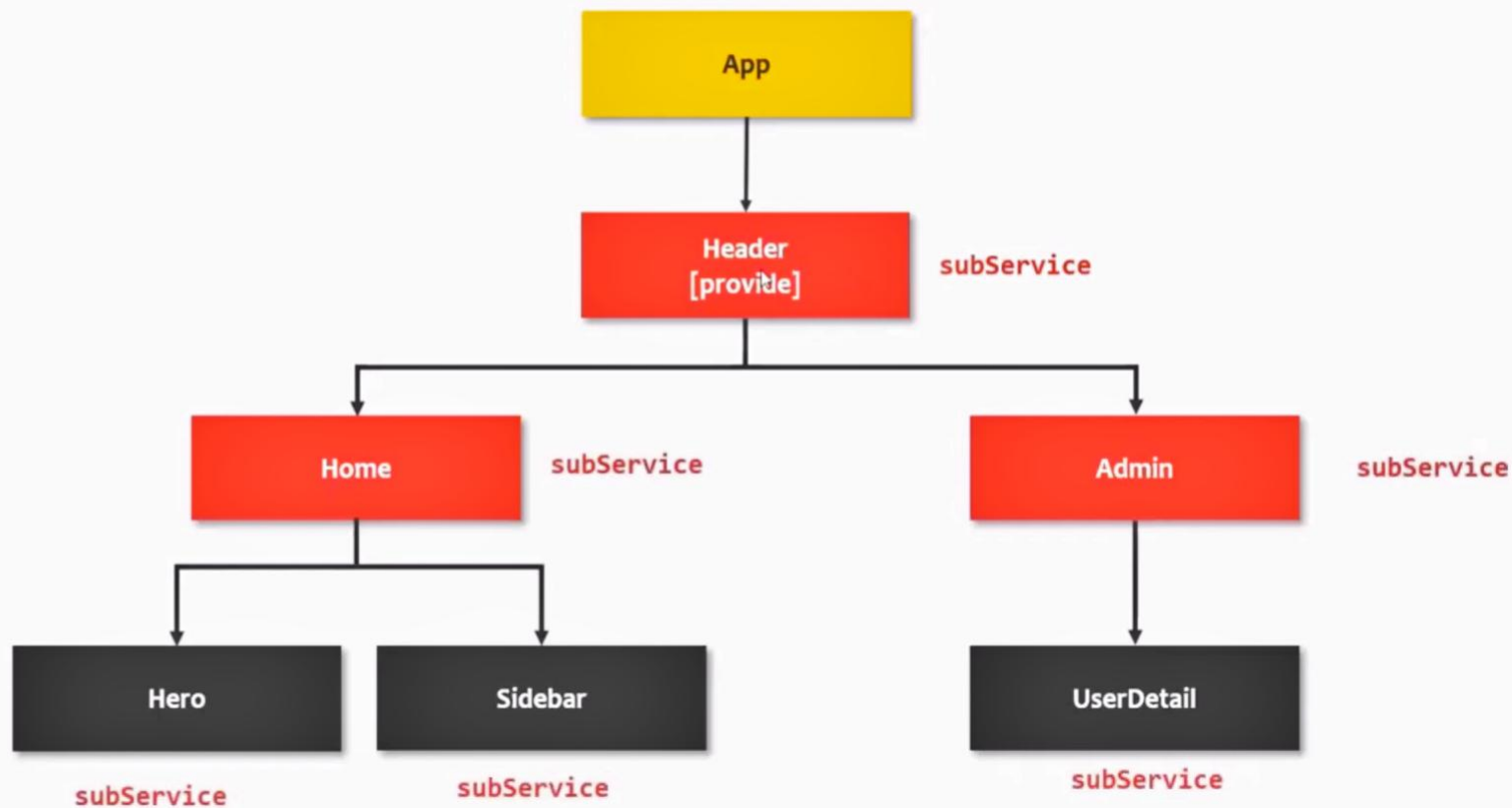


Hierarchical Injection

When we provide a dependency on a component, the same instance of that dependency is injected in component class and all its child components and their child components. This is called as **hierarchical injection**.







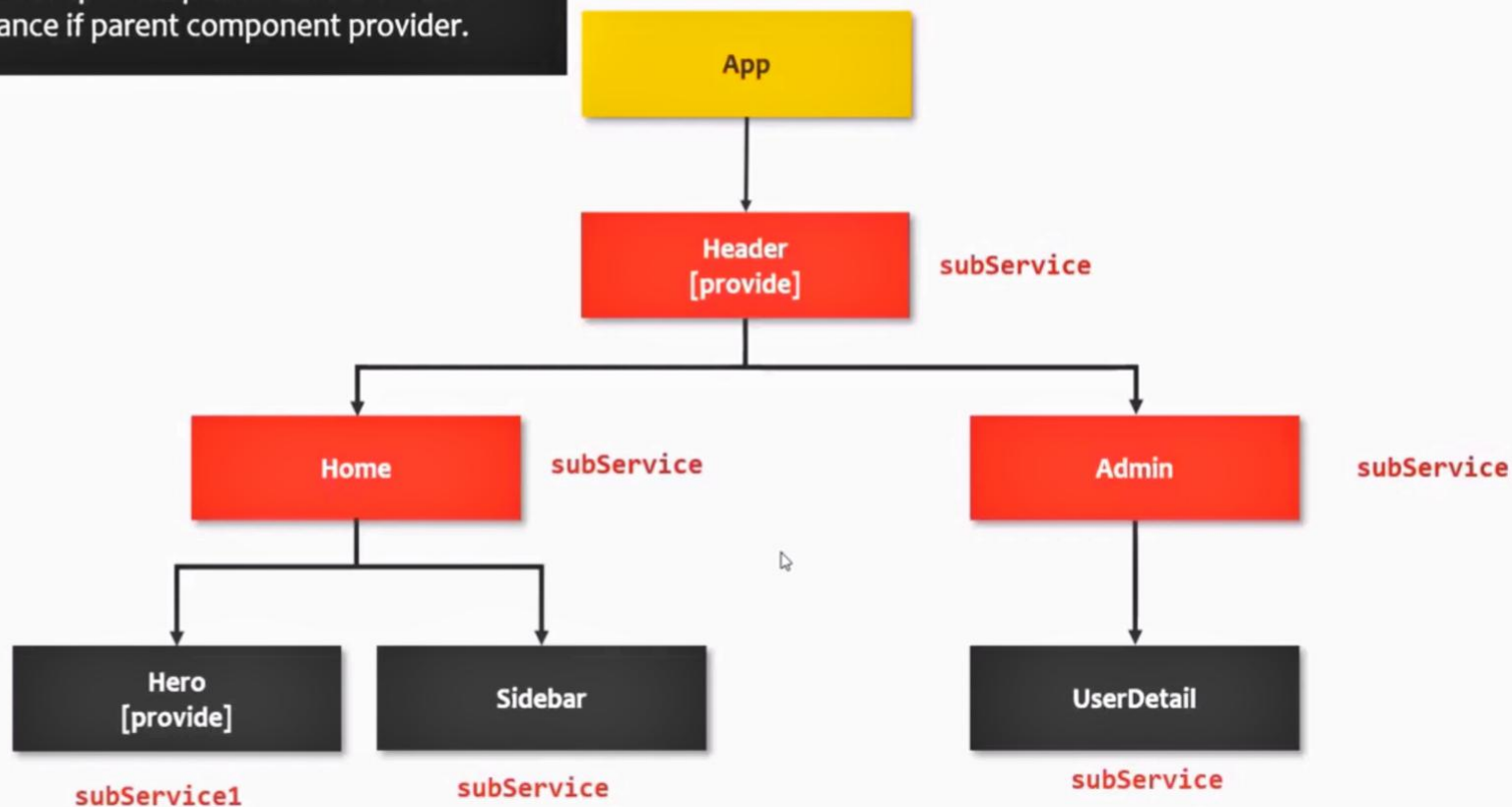
Dependency Override

When we provide a dependency on a component and we also provide a dependency on its child component, child component dependency instance will override its parent component dependency instance.

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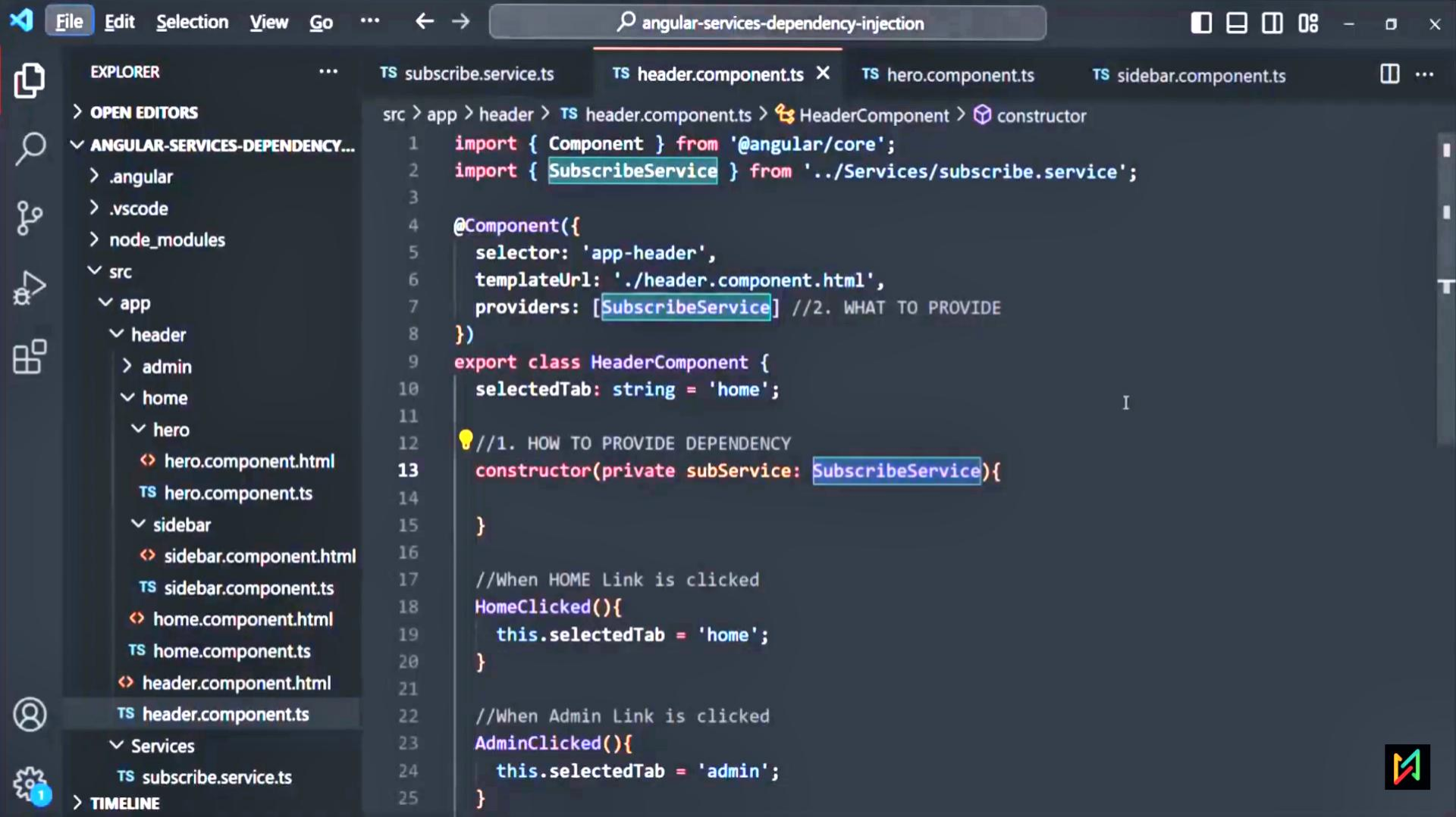
3 Child component provider overrides the instance if parent component provider.

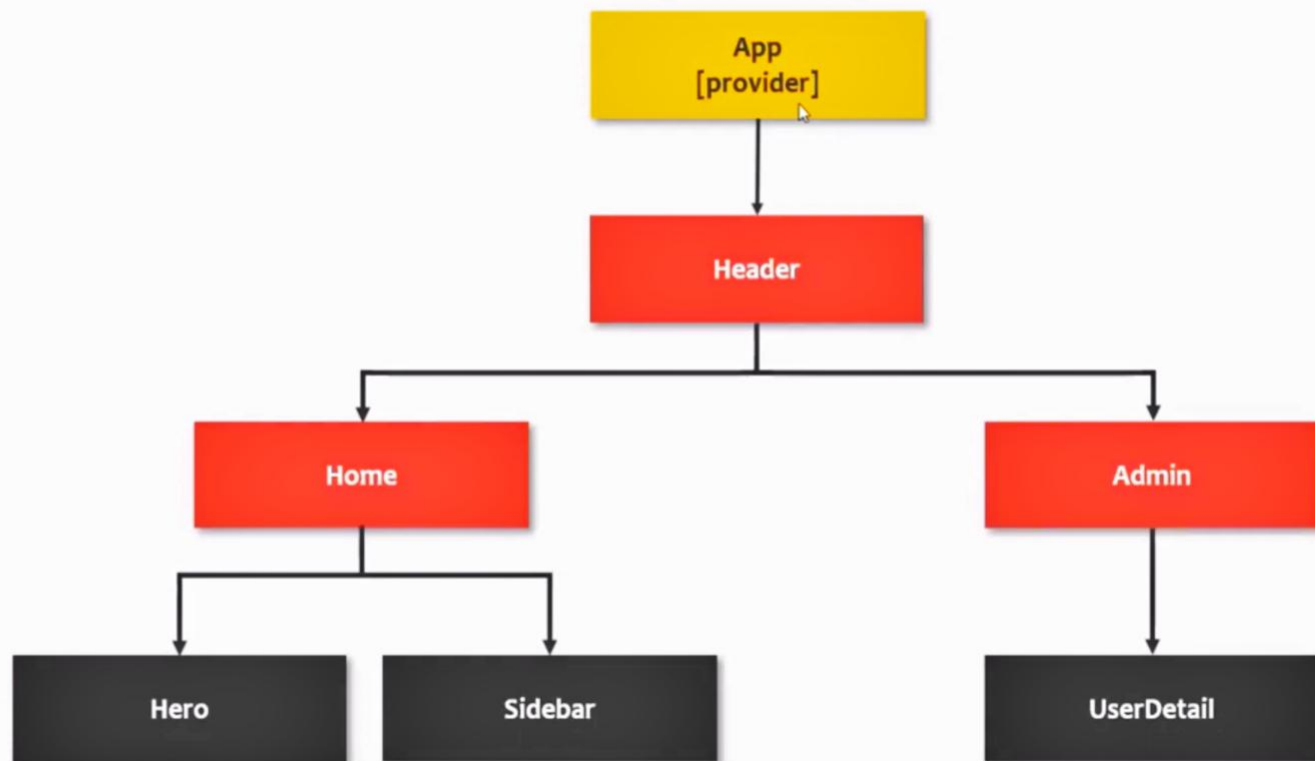


Dependency injection on Root Component

When we provide a dependency on root component, same instance of that dependency is injected to all components, directives and services.



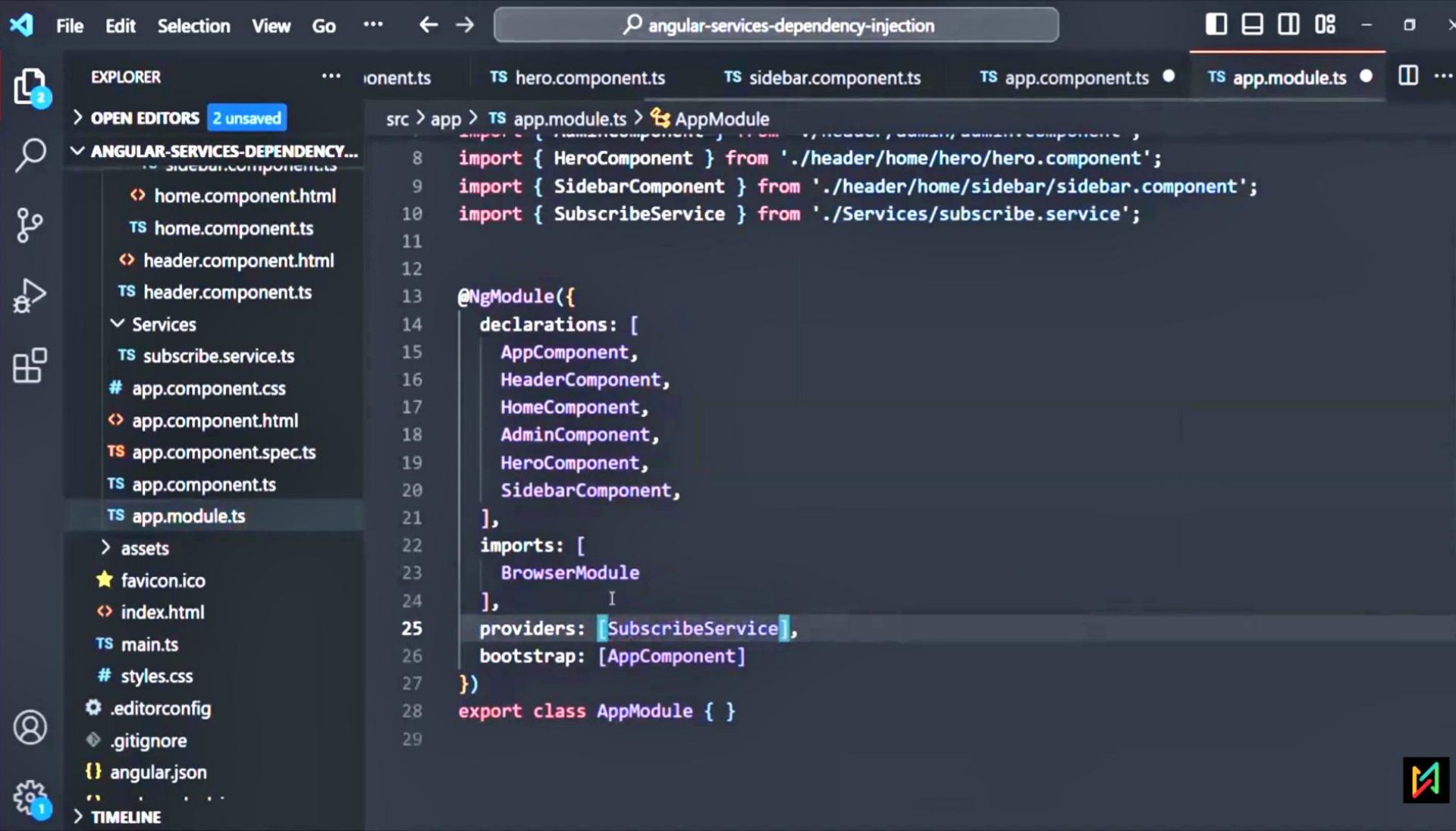




Module Injector

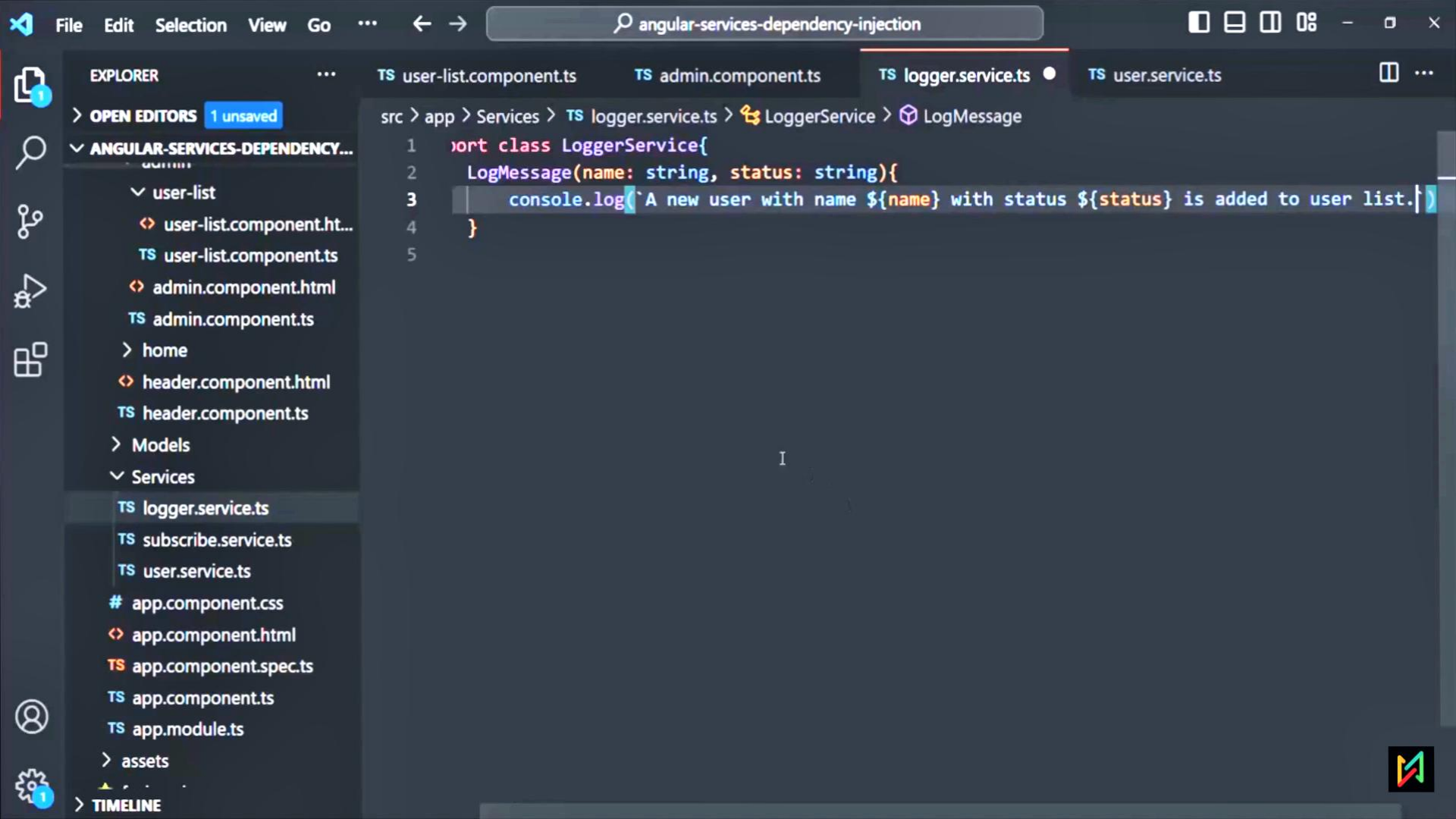
We can also inject a service from Module class. In that case same instance of the dependency will be available throughout the Angular application. In this way we implement singleton pattern where a single instance is shared throughout the application.

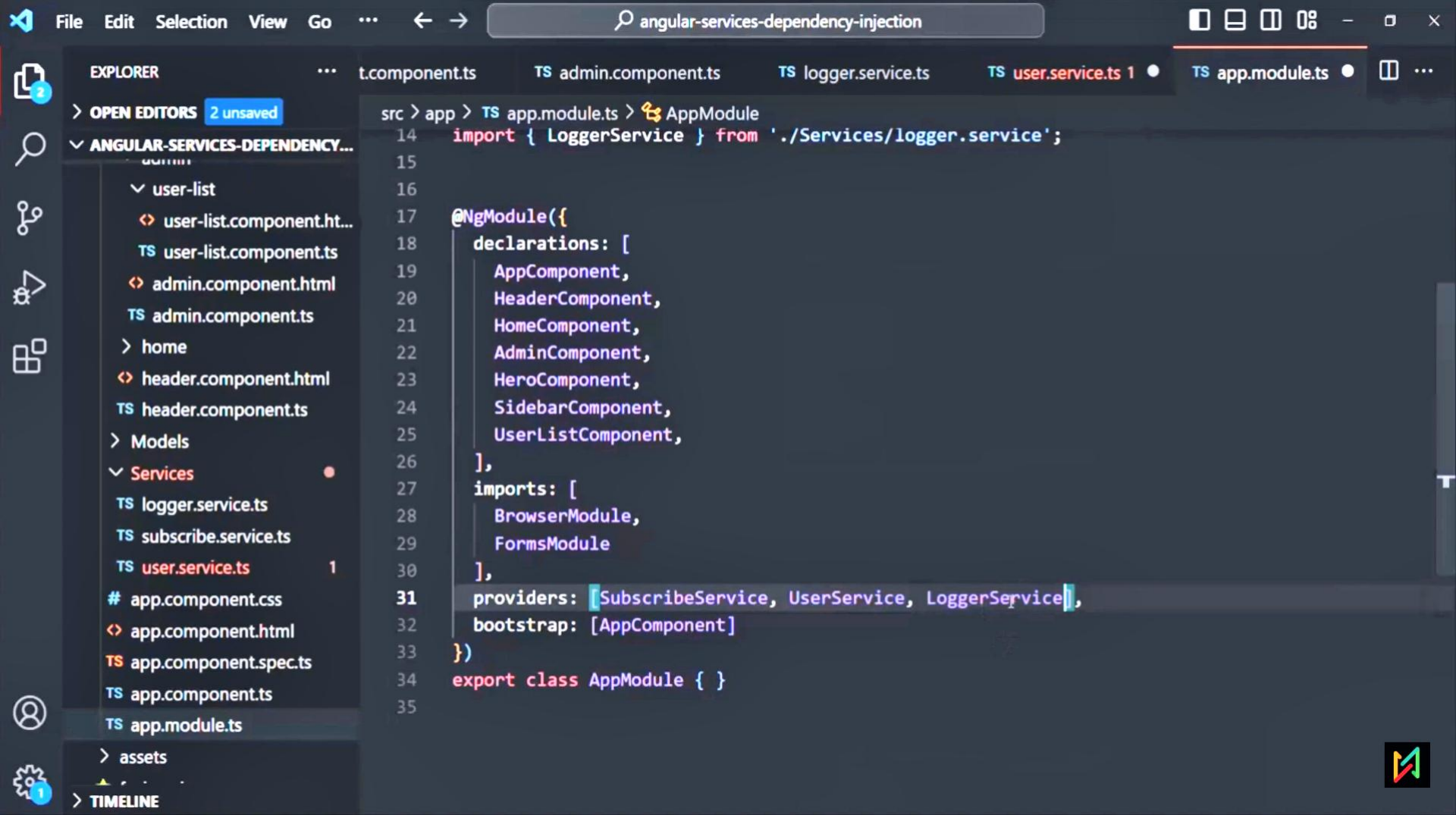


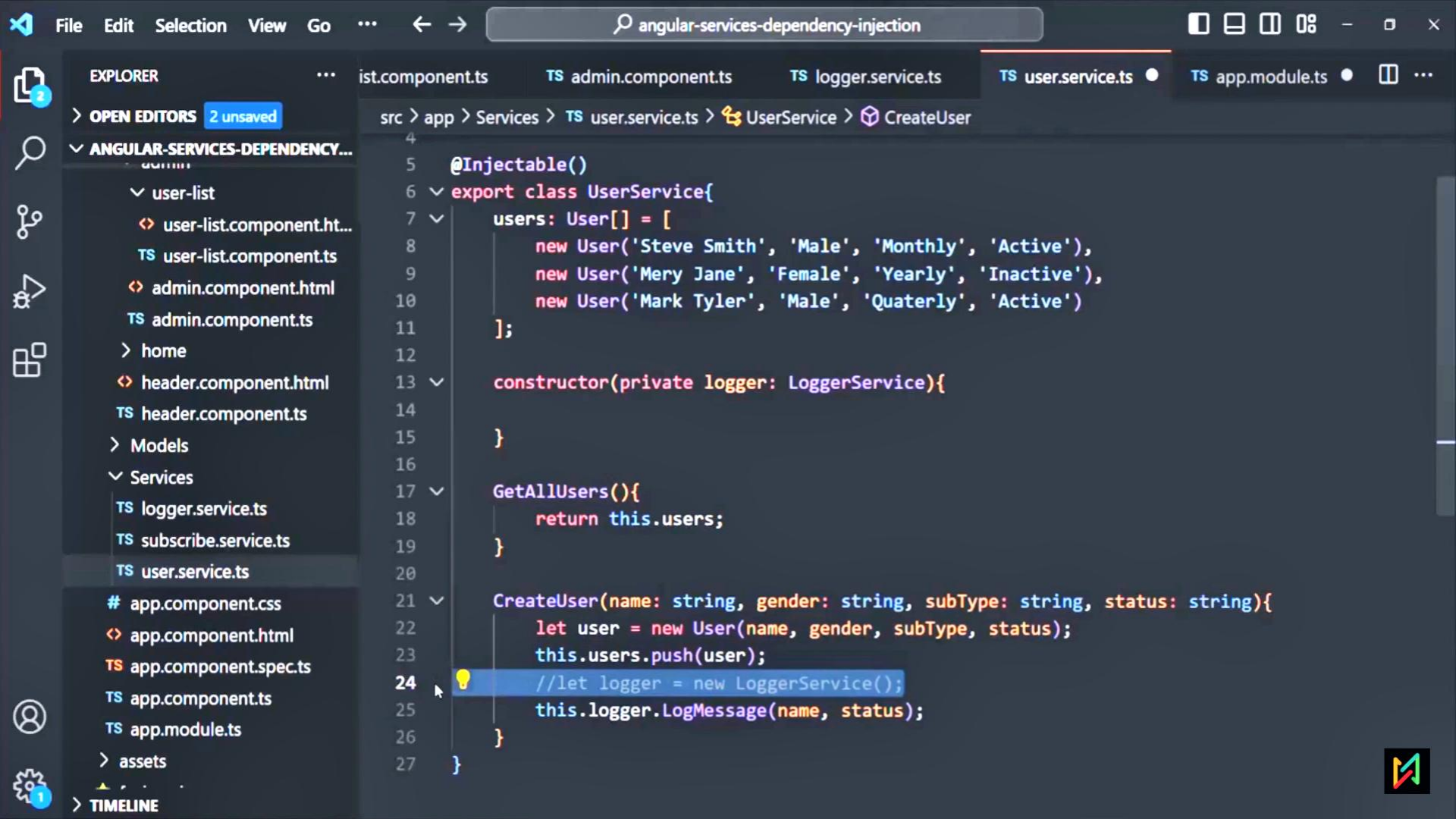


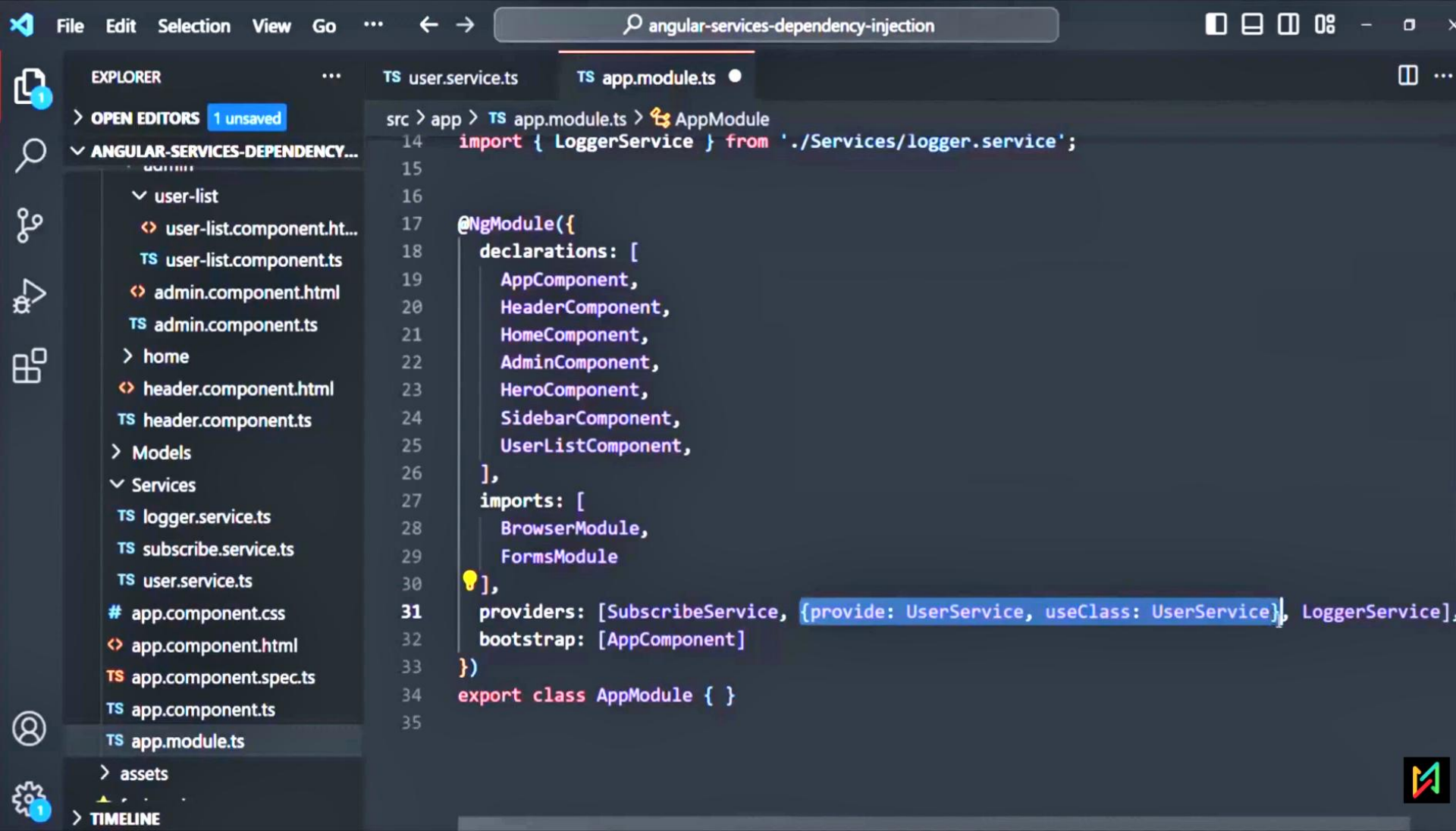
Injecting Service into Another Service











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▼ ANGULAR-SERVICES-DEPENDENCY...

▼ user-list

user-list.component.html

TS user-list.component.ts

admin.component.html

TS admin.component.ts

> home

header.component.html

TS header.component.ts

> Models

▼ Services

TS logger.service.ts

TS subscribe.service.ts

TS user.service.ts

app.component.css

app.component.html

TS app.component.spec.ts

TS app.component.ts

TS app.module.ts

> assets

> TIMELINE

TS user.service.ts

TS app.module.ts

TS user-list.component.ts

src > app > TS app.module.ts > AppModule

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import { LoggerService } from './Services/logger.service';

@NgModule({

declarations: [

AppComponent,

HeaderComponent,

HomeComponent,

AdminComponent,

HeroComponent,

SidebarComponent,

UserListComponent,

],

imports: [

BrowserModule,

FormsModule

],

providers: [

SubscribeService,

provide: 'USER_SERVICE', useClass: UserService,

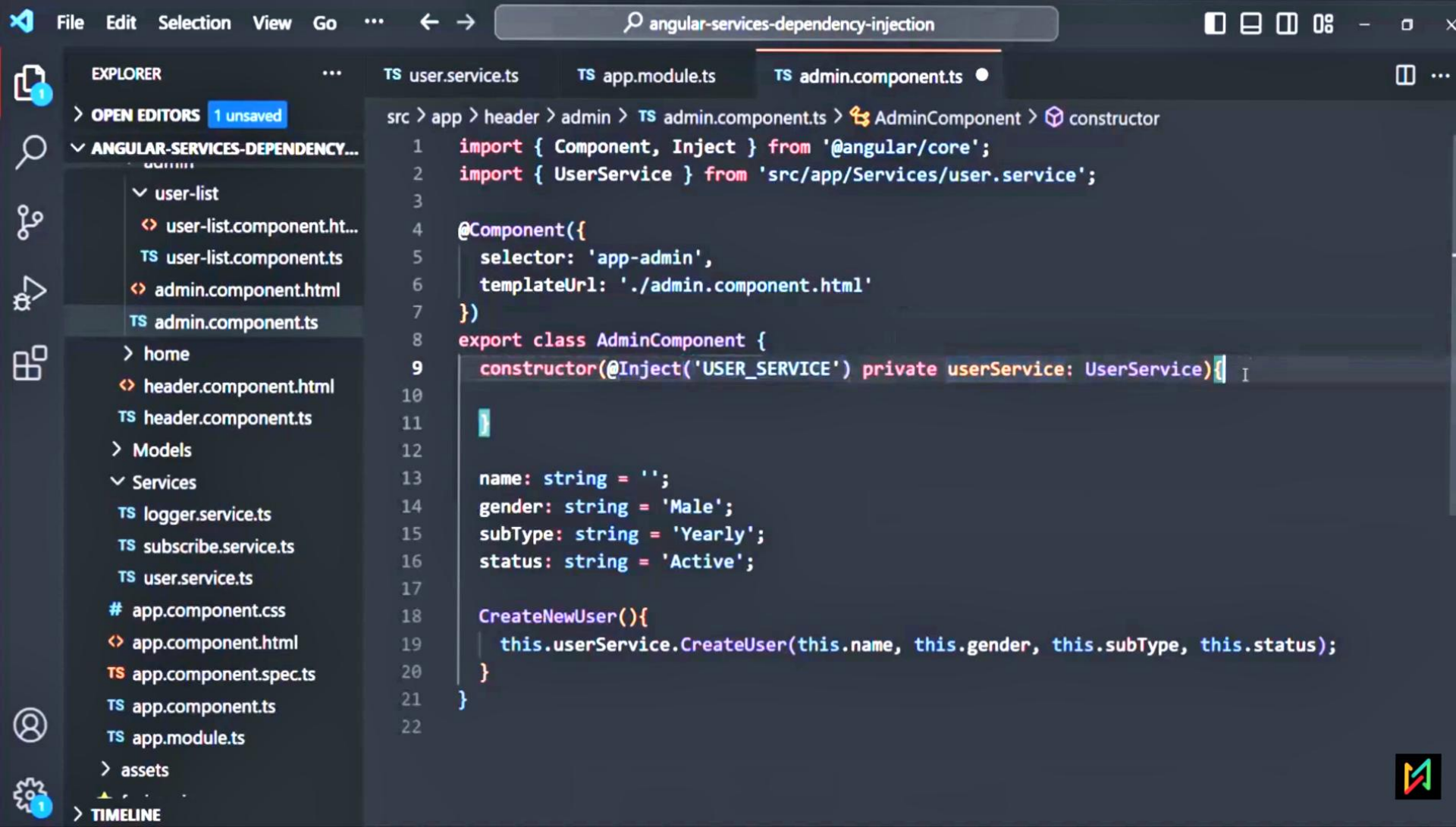
LoggerService,

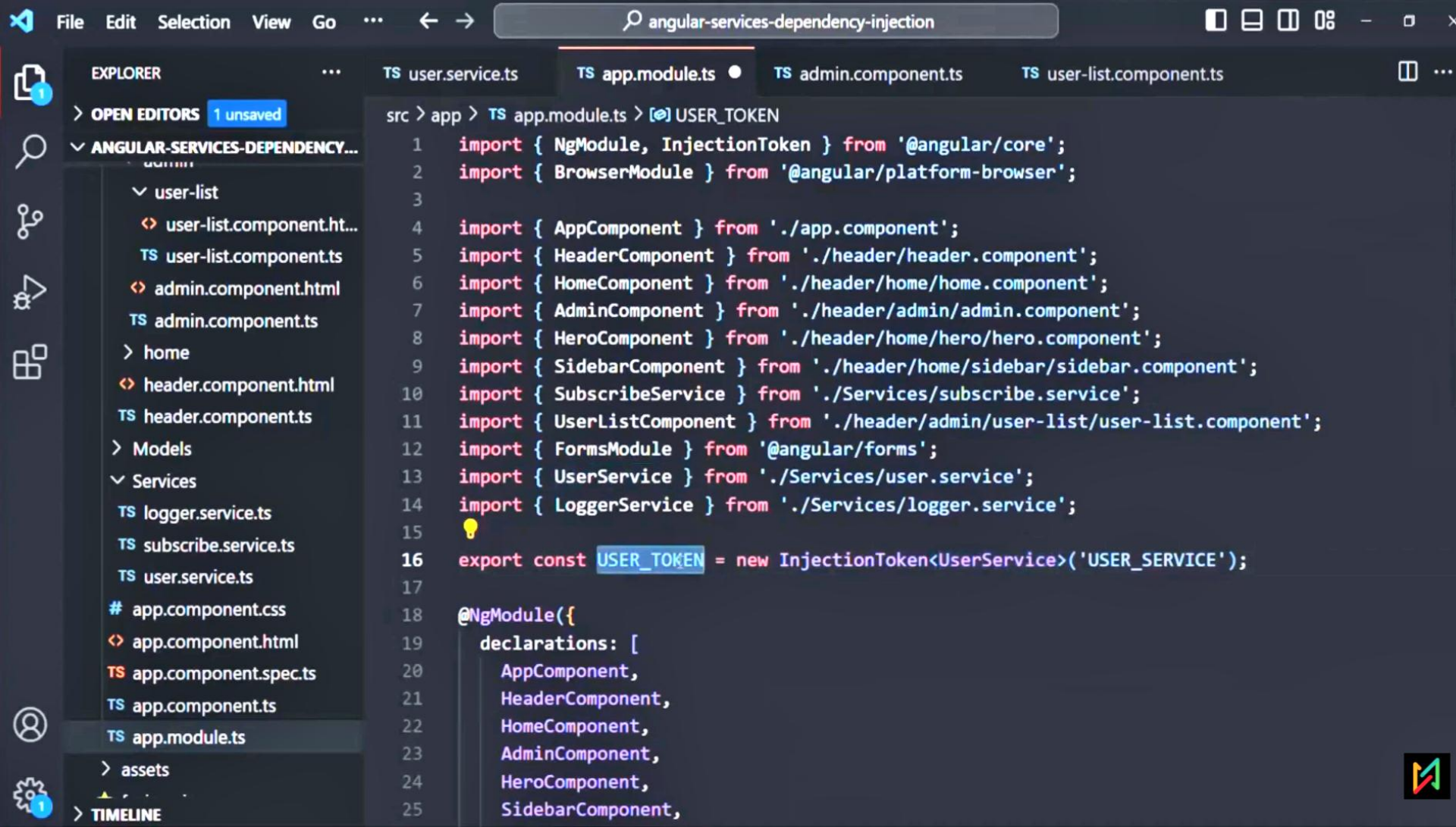
],

bootstrap: [AppComponent]

})

export class AppModule { }





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EXPLORER ... TS user.service.ts TS app.module.ts TS admin.component.ts TS user-list.component.ts

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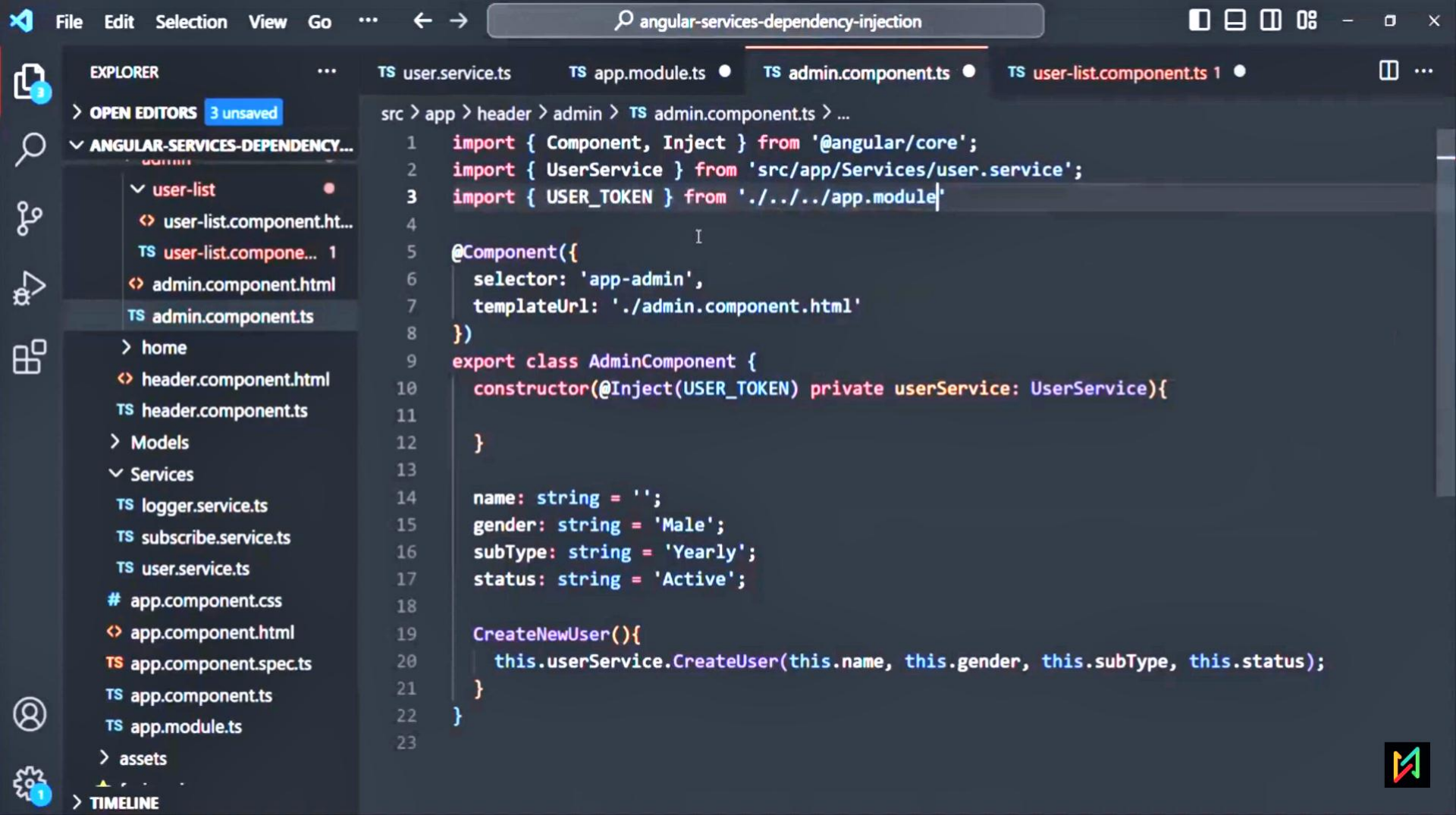
ANGULAR-SERVICES-DEPENDENCY...

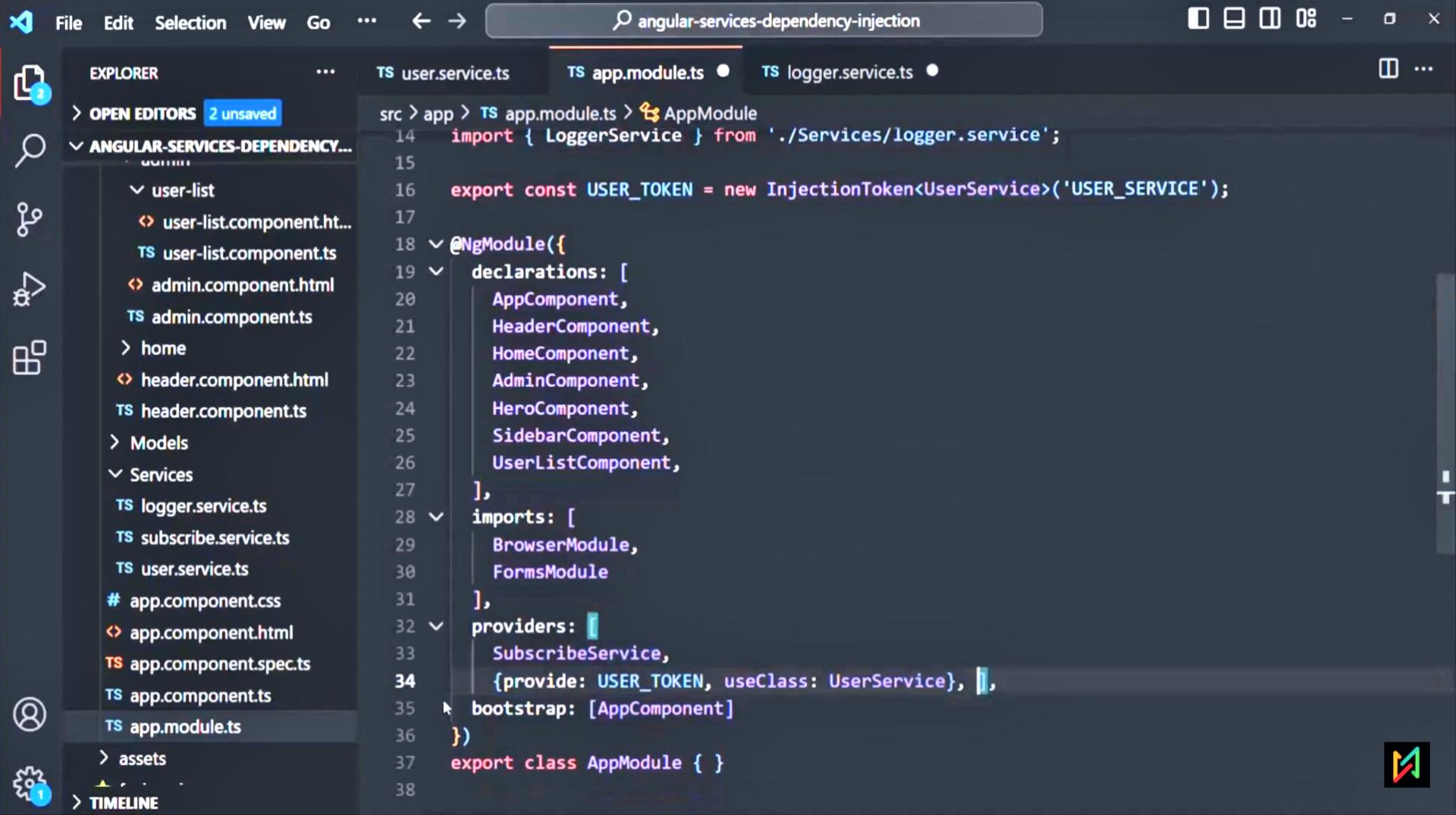
- user-list
 - user-list.component.html
 - TS user-list.component.ts
- admin.component.html
- TS admin.component.ts
- home
- header.component.html
- TS header.component.ts
- Models
- Services
 - TS logger.service.ts
 - TS subscribe.service.ts
 - TS user.service.ts
- # app.component.css
- app.component.html
- TS app.component.spec.ts
- TS app.component.ts
- TS app.module.ts
- assets

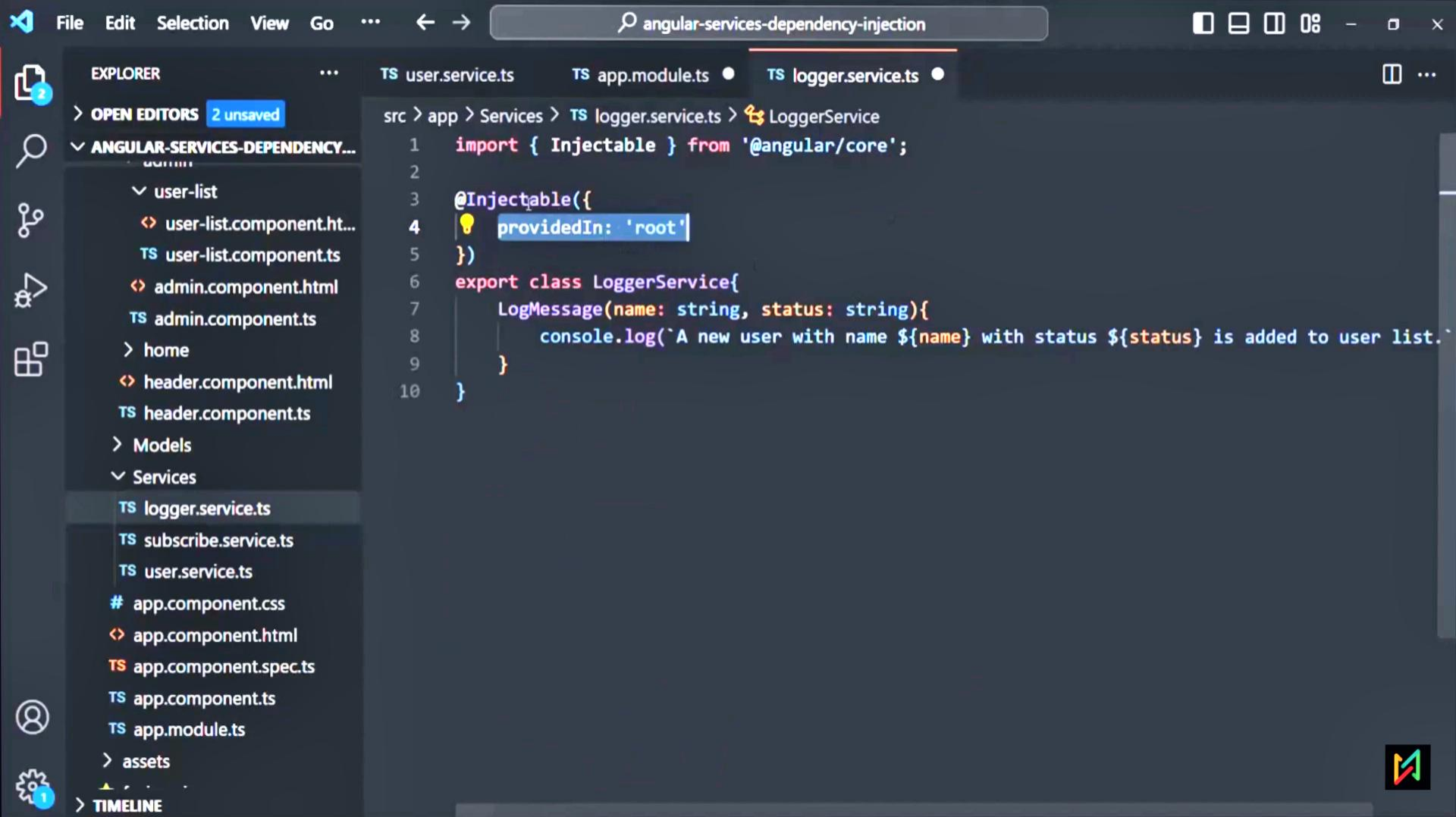
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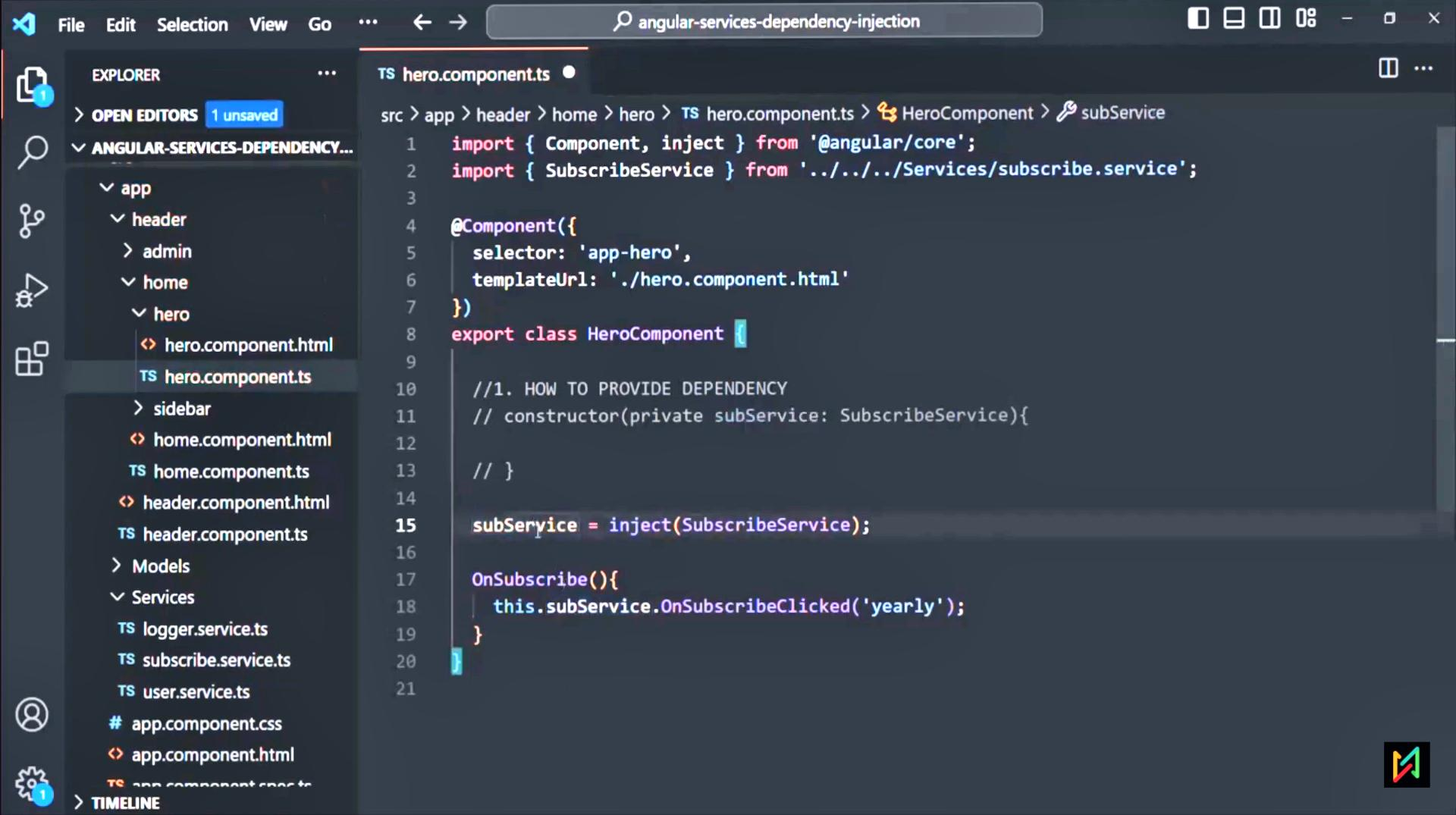
src > app > TS app.module.ts > AppModule

```
18 @NgModule({
19   declarations: [
20     AppComponent,
21     HeaderComponent,
22     HomeComponent,
23     AdminComponent,
24     HeroComponent,
25     SidebarComponent,
26     UserListComponent,
27   ],
28   imports: [
29     BrowserModule,
30     FormsModule
31   ],
32   providers: [
33     SubscribeService,
34     {provide: USER_TOKEN, useClass: UserService},
35     LoggerService],
36   bootstrap: [AppComponent]
37 })
38 export class AppModule { }
39
```



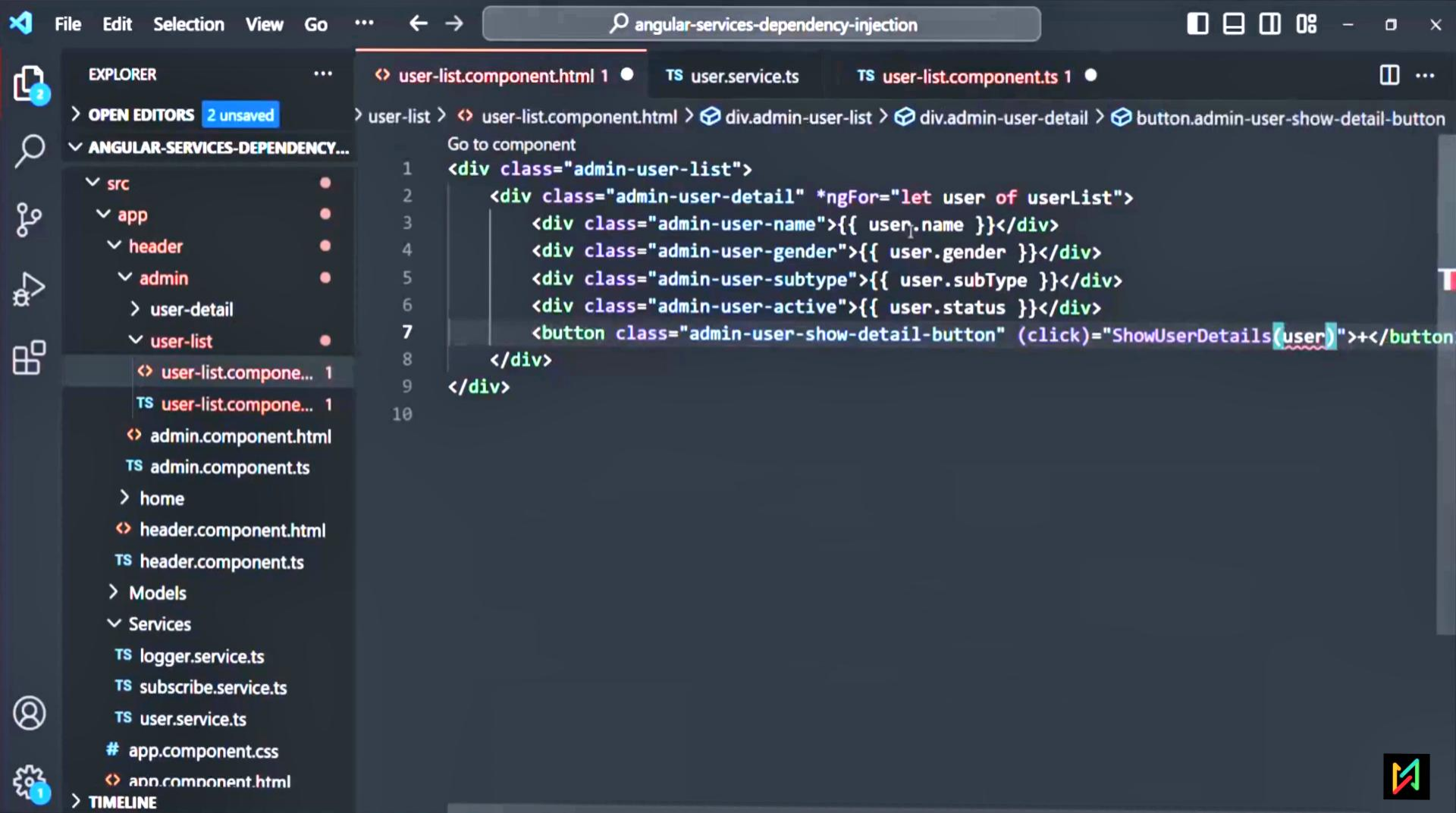


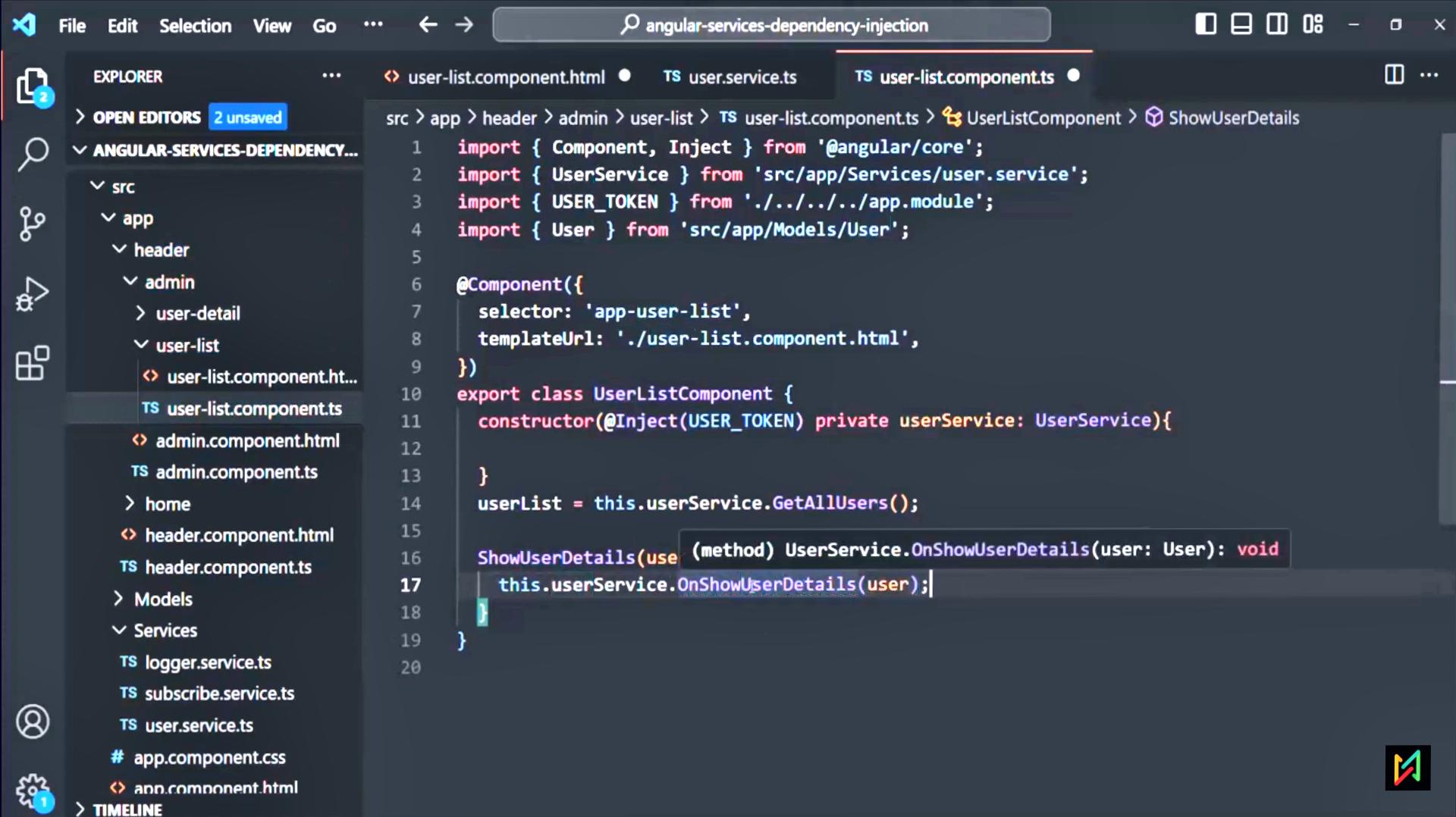




Component Interaction using Services







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EXPLORES

...

user-list.component.html

TS user.service.ts

TS user-list.component.ts

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ANGULAR-SERVICES-DEPENDENCY-...

src

app

header

admin

user-detail

user-list

user-list.component.ht...

TS user-list.component.ts

admin.component.html

TS admin.component.ts

home

header.component.html

TS header.component.ts

Models

Services

logger.service.ts

subscribe.service.ts

user.service.ts

app.component.css

app.component.html

TIMELINE

src > app > Services > TS user.service.ts > UserService > OnShowUserDetails

1 import { Injectable, EventEmitter } from '@angular/core';

2 import { User } from "../Models/User";

3 import { LoggerService } from "../logger.service";

4

5 @Injectable()

6 export class UserService{

7 users: User[] = [

8 new User('Steve Smith', 'Male', 'Monthly', 'Active'),

9 new User('Mery Jane', 'Female', 'Yearly', 'Inactive'),

10 new User('Mark Tyler', 'Male', 'Quarterly', 'Active')

11];

12

13 constructor(private logger: LoggerService){

14

15 }

16

17 OnUserDetailsClicked: EventEmitter<User> = new EventEmitter<User>();

18

19 OnShowUserDetails(user: User){

20 this.OnUserDetailsClicked.emit(user);

21 }

22

23 GetAllUsers(){

24 return this.users;

25 }

